

SEYOUNG JOUNG

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AREAS OF FOCUS

Human-AI Interaction for Inclusive Technology & Sensory Augmentation

- **Expertise:** Computer Vision (3D & Depth), Deep Learning, Machine Learning
- **Interests:** Human-Centered Spatial Design (Virtual & Physical), Human-AI Interaction

EDUCATION

Arizona State University (ASU), Arizona, USA

- M.S. in Media Arts and Sciences

2026-

Ewha Womans University (EWU), Seoul, Korea

- M.S. in Artificial Intelligence and Software ([Computer Vision Lab](#))
- B.S. in Statistics, B.S.E. in Computer Science and Engineering

2022–2025

2017–2022

PUBLICATIONS AND PRESENTATIONS

1. J. Kim, J. Yoon, **S. Joung**, J. Roh, and Y. Lee, “Development and Application of VR Sports Experience Program for Middle School Students with Development Disabilities,” in *Proceedings of the Korea Digital Contents Society Summer Conference, No. 18*, Jul. 2026.
 - Developed a 3-stage VR sports program across 6 disciplines for students with developmental disabilities, verifying its content validity and baseline efficacy to earn the **Best Paper Award**.
2. H. Choi, H. Lee, **S. Joung**, H. Park, J. Kim, and D. Min, “Emerging property of masked token for effective pre-training,” in *European Conference on Computer Vision (ECCV)*, Sep. 2024.
 - Theoretically proved the convergence of masked token attention toward visible patches by minimizing row-wise softmax entropy via mathematical induction, establishing the foundational evidence for the pre-training efficiency of Masked Token Optimization (full proof published in the Supplementary Material).
3. **S. Joung**, “DeCoMon: Depth completion-guided monocular depth estimation via knowledge distillation,” Master’s thesis, Ewha Womans University, Seoul, Republic of Korea, 2024.
 - Developed *DeCoMon*, a novel Knowledge Distillation framework that resolves scale ambiguity in monocular depth estimation utilizing a Depth Completion teacher model, with results showing superior zero-shot generalization on KITTI and Make3D benchmarks for LiDAR-free, scale-consistent depth prediction.
4. S. Lim, **S. Joung**, and A. Duran, “U.S. women’s colleges through the ages: A systematic review of the literature using BERTopic as a research approach,” poster presentation (accepted, April 2026), *Arizona State University Institute for Social Science Research Graduate Student Poster Contest*, AZ, USA.
 - Invited as a coauthor to develop BERTopic extension of a paper presented at 2025 meeting of a top-tier U.S. higher education conference (Association for the Study of Higher Education, acceptance rate $\approx 30\%$).
 - Manuscript in preparation; Targeting *Review of Educational Research* (ranked 2/762 education journals by 5-year impact factor) for submission, designing and implementing a complete BERTopic analysis pipeline.

TECHNICAL SKILLS

Languages & Tools

Python, C/C++, Java, SQL, Git/GitHub, Linux (Ubuntu)

AI/ML & Computer Vision

PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Weights & Biases (W&B)

Graphics & Video Media

Adobe Illustrator, Premiere Pro

Research & Analysis

R, SAS, Statistical Modeling, Mathematical Derivation & Formal Proofs

AI-Assisted Development

Prompt Engineering, Cursor-based Rapid Prototyping

RESEARCH AND LEADERSHIP PROJECTS

- Graduate Research Assistant**, LG Electronics Industry-Academia Collaboration **2023**
Self-Supervised Learning for Driving Error and Obstructed Zone Prediction in Robot Cleaner (PI: Dr. Dongbo Min)
- Defined failure case taxonomies for autonomous navigation and curated high-fidelity datasets using mobile robots to establish the foundational framework for self-supervised error detection.
- Lead Developer for Natural Language Processing**, EWU Computer Science Capstone Project **2021–2022**
AWESUM: AI-Based Automatic Lecture Video Summarization Service. [\[code\]](#)
- Architected an AI-driven learning assistant, *AWESUM*, by integrating BERT and T5 models to automate lecture summarization and knowledge assessment to bridge the gap between digital content and learner comprehension.
- Undergraduate Research Assistant**, EWU College of Education **2020**
Comparison of Education Policies on Digital Literacy Education (PI: Dr. Hyounjin Ok)
- Conducted qualitative interviews with eight Korean participants aged 65+ to investigate the digital divide, and synthesized insights to foster digital empowerment and inclusivity for the elderly.
- President of College Student Council**, EWU College of Natural Sciences **2019**
Barrier-Free College-Wide Annual Cultural Events
- Ensured multi-dimensional accessibility by providing real-time captioning for audiovisual content (Freshman Orientation) and constructing barrier-free pathways for film venues (Inclusive Rights Advocacy Film Festival).

PROFESSIONAL ACTIVITIES

- Project Lead**, BITAmin 6th cohort [\[link\]](#) (Major Multi-University Data Analytics Union, Korea) **2020–2021**
- Enhanced personalized fashion discovery by developing a content-based recommendation system that utilized CNN-derived feature vectors and average RGB values to overcome the Cold Start problem in traditional collaborative filtering. [\[code\]](#)
 - Recruited through a highly selective process (12:1 admission rate); led ML/DL study groups and weekly technical seminars to foster a collaborative research environment while achieving top-tier competition results.
- Fellow**, DAITGIRLS (Sponsored by the Ministry of Science and ICT, Korea) **2019**
- Completed an intensive 3-month data science residency, mastering full-stack data literacy ranging from statistical theory and computer architecture to collaborative version control using Git/GitHub.

TEACHING EXPERIENCE

- Graduate Teaching Assistant**, College of Artificial Intelligence, EWU **2022–2024**
- Data Structure (Fall 2022, Fall 2024)
 - Foundation of Artificial Intelligence (Spring 2023, Fall 2023, Spring 2024)
- Python Peer Mentoring**, EWU **2022**
- Developed personalized Python curriculum and workbooks to scaffold computational thinking for students.
- Undergraduate Teaching Assistant**, College of Engineering, EWU **2020**
- Problem Solving and Software Programming (Fall 2020)

ATHLETICS

- Elite Player (No. 17)**, FS Goyang Ladies [\[Article\]](#) **2023–2025**
- Club Founder and Captain (No. 10)**, FS PRIME [\[Video 1\]](#) [\[Video 2\]](#) [\[Video 3\]](#) **2021–2024**
- Player (No. 13)**, FC QOK, EWU **2017–2019**

AWARDS AND HONORS

- Best Paper Award**, Korea Digital Contents Society Summer Conference (DCS 2026) **Jul 2026**
- Ewha-Boeing Scholarship for CS Graduate Students**, Boeing (Partial Tuition) **Fall 2024**
- Merit-Based STEM Graduate School Admission Scholarship**, EWU (1 Year Full Tuition) **2023**
- Merit-Based Scholarship**, Institute for Professional Women in Globalization Era **Fall 2023**
- Undergraduate Teaching Assistantship Scholarship**, EWU (Partial Tuition) **Fall 2020**
- Undergraduate Research Assistantship Scholarship**, EWU (Partial Tuition) **Fall 2020**